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Governance, structure, and access: a Bayesian hierarchical analysis of water, sanitation, and hygiene in South Asia

Mohammad Naim Azimi^{1*} and Mohammad Mafizur Rahman¹

Abstract

Background Access to safe water, sanitation, and hygiene (WASH) remains inadequate for large populations in South Asia, constrained by persistent structural inequalities, institutional weaknesses, and uneven development. This study investigates the complex interplay of structural, demographic, environmental, and institutional factors shaping WASH outcomes across eight South Asian countries over the period 2000–2024. Employing a Bayesian hierarchical linear model—robustly validated through alternative specifications—we provide a comprehensive probabilistic assessment of WASH outcomes. A key methodological contribution is the construction of a composite governance quality index using an autoencoder neural network based on six governance dimensions: voice and accountability, regulatory quality, political stability, control of corruption, government effectiveness, and rule of law.

Results Urbanisation improves sanitation access and reduces open defecation but strains water access, highlighting its dualistic nature. Health expenditures consistently enhance WASH outcomes. Economic capacity, sectoral vulnerability, precipitation shocks, and governance exhibit no direct impact. Education significantly improves water access and reduces open defecation. Population density exerts small but positive effects, suggesting economies of scale in service delivery. Digitalisation also emerges as a key driver of WASH outcomes, reflecting the growing role of information technology in service delivery and accountability mechanisms. A counterintuitive positive association between inequality and WASH outcomes suggests Kuznets-style transitional dynamics, where benefits initially accrue to more affluent groups. However, these relationships shift notably when governance is modelled as a moderator. Governance quality acts as a key moderating mechanism, amplifying the benefits of growth, urbanisation, and health spending while reshaping the influence of education and inequality.

Conclusions Governance does not operate primarily as a direct driver of WASH outcomes. Instead, it acts as a critical enabling mechanism that shapes how structural factors translate into equitable access. Strengthening governance quality amplifies the effectiveness of sectoral investments and urbanisation policies—a finding with important implications for WASH policy and institutional reform across South Asia.

Keywords Water, Sanitation, Open defecation, South Asia, Governance

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Introduction

The global pursuit of Sustainable Development Goal (SDG) 6—to ensure universal access to water, sanitation, and hygiene (WASH)—is alarmingly off track, with far-reaching implications for human health, productivity, and socioeconomic development [1, 2]. Despite two decades of international commitment and considerable public investment, over 2.2 billion people still lack safely managed drinking water services [3]. The health burden is substantial: diarrheal disease remains the second leading cause of death among children under five globally, with unsafe WASH conditions accounting for an estimated 72% of diarrheal deaths in this group [4]. Recent evidence shows that improving water quality and access is associated with significant reductions in all-cause childhood mortality, although multi-component WASH interventions do not always generate the synergistic effects previously assumed [5]. Beyond enteric infections, inadequate WASH infrastructure is increasingly recognised as a driver of antimicrobial resistance, with poor sanitation and hygiene accelerating the transmission of resistant pathogens in low- and middle-income countries [6]. Inadequate WASH conditions contribute to more than 830,000 preventable deaths from waterborne diseases annually, disproportionately affecting children under five [7–9]. Beyond health, limited WASH access constrains educational attainment, labour participation, and gender equality, while heightening vulnerability to environmental and economic shocks [10]. These persistent shortfalls—particularly acute in low- and lower-middle-income countries—suggest that rising income and infrastructure investments have not reliably translated into universal, equitable, and sustainable service provision. As the 2030 SDG deadline approaches, closing this gap has become both a humanitarian priority and a strategic development objective for global development and climate resilience [11].

South Asia exemplifies the disconnect between investment and outcomes. The region experienced sustained economic growth averaging more than 5% annually over the two decades prior to the COVID-19 pandemic [12], alongside significant public investment in WASH initiatives, including India's Swachh Bharat Mission, Bangladesh's Sector Development Plan, and Nepal's Total Sanitation Campaign [13–15]. Despite these advances, access remains highly uneven. Approximately 134 million people still lack basic drinking water services, while more than 600 million lack safely managed sanitation, with rural coverage lagging urban areas by 20–30% points [16]. These aggregate statistics conceal significant intra-national disparities shaped by income, gender, and social stratification [11]. Rapid urbanisation and population density further strain infrastructure, intensifying challenges such as groundwater depletion and contamination.

The coexistence of strong economic dynamism with persistent service fragmentation highlights the limitations of growth-centric explanations and underscores the need for a more integrated analytical framework that accounts for institutional and governance dynamics.

A critical limitation persists within the WASH literature: an overreliance on supply-side and resource-based approaches that inadequately explain divergent outcomes across similarly endowed countries [9, 17–22]. This supply-side bias manifests in several ways. Studies frequently treat infrastructure construction, capital expenditure, and technological transfer as sufficient conditions for improved outcomes, while systematically underweighting the institutional, behavioural, and governance factors that determine whether built infrastructure is used, maintained, and sustained [23, 24]. Large-scale evaluations of rural water supply programmes in Sub-Saharan Africa and South Asia have documented high rates of infrastructure non-functionality—with estimates ranging from 30 to 60% of rural water points non-operational within two to five years of construction—attributable primarily to the absence of post-construction support systems and accountability mechanisms rather than to technical or financial shortfalls [25]. Similarly, systematic reviews of sanitation promotion programmes find that latrine construction subsidies consistently fail to generate sustained usage in the absence of complementary demand-side and institutional interventions [23, 26]. While economic growth and infrastructure expansion are necessary conditions for improving WASH access, they do not guarantee effective service delivery or sustained usage. Empirical studies increasingly show that improvements in infrastructure alone often fail to translate into improved outcomes in the absence of strong institutional capacity and governance [27–29]. The prevailing focus on aggregate access and infrastructure indicators therefore reflects a broader supply-side bias that overlooks the mechanisms through which resources are allocated, implemented and sustained—and this conceptual gap directly connects to the fragmented state of the empirical evidence.

The existing evidence base analyses economic, social, demographic, or institutional determinants largely in isolation, offering limited insights into how these factors interact to shape WASH outcomes. While some contributions incorporate governance—such as Achuo and Asongu [30], who identify threshold effects in the interaction between public spending and governance—these analyses do not fully explain the mechanisms underlying such interactions or how they vary across institutional settings. Likewise, qualitative studies [31] emphasise the interconnected nature of financial, institutional, environmental, and social drivers but often lack empirically testable frameworks. As a result, governance

is typically treated as an additive input rather than as a structural mechanism that conditions the effectiveness of other determinants. Empirical models explicitly capturing these interaction effects remain scarce, particularly in South Asia, where institutional heterogeneity is substantial. This gap limits the explanatory power of existing studies and contributes to policy prescriptions that emphasise resource expansion without adequately addressing institutional capacity.

Building on these limitations, this study seeks to advance a more comprehensive understanding of the determinants of WASH outcomes in South Asia. Specifically, it addresses the following research questions: (a) To what extent do economic, demographic, social, and environmental factors directly influence WASH outcomes in South Asia? (b) How does governance quality condition the effectiveness of these structural drivers? (c) Do governance moderation effects differ across sanitation access, water access, and open defecation outcomes? Addressing these questions is critical not only for advancing the theoretical and empirical frameworks underpinning WASH research but also for enabling evidence-based policymaking and designing institutionally responsive WASH strategies that accelerate equitable and sustainable progress toward SDG 6 across South Asia.

This paper aims to advance the literature through a series of novel theoretical, methodological, and empirical contributions. Theoretically, it reconceptualises governance not as a direct input but as a structural effect modifier that conditions how economic, demographic, and fiscal drivers translate into WASH outcomes. This framing moves the debate beyond whether resources matter to explaining why identical resource endowments yield divergent outcomes across institutional contexts, thereby clarifying the mixed evidence surrounding WASH interventions. Methodologically, the study employs a Bayesian Hierarchical Linear Model (HLM) applied to a South Asian dataset, enabling the modelling of unobserved heterogeneity and parameter variations across countries, producing results that are both statistically rigorous and policy relevant. In addition, the study develops a context-sensitive governance index using an autoencoder-based multidimensional reduction approach, which extracts latent governance signals from multiple institutional indicators while minimising information loss—improving upon the use of generic global composite indices. Empirically, the analysis addresses the persistent disconnect between resource availability and service outcomes by demonstrating that governance acts as a key transmission mechanism through which structural capacity is converted into effective WASH provision.

The remainder of this study is structured as follows. [Section “Literature review”](#) reviews the theoretical and empirical literature, leading to the development of

new testable hypotheses. [Section “Data and variables”](#) describes the data and variables. [Section “Estimation methods”](#) outlines the empirical methodology. [Section “Statistical results”](#) presents the results, followed by a discussion in [Sect. “Discussion”](#). [Section “Conclusions”](#) concludes with policy implications and limitations.

Literature review

Theoretical background

The pursuit of universal access to water, sanitation, and hygiene (WASH) is fundamentally a question of how societal resources are mobilised and allocated—a problem addressed in several interconnected theoretical traditions. Welfare economics provides the foundational justification for state intervention, framing WASH services as public and merit goods whose positive externalities and essential role in human welfare lead to chronic undersupply under pure market conditions [32, 33]. Even when private demand exists, consumers tend to undervalue social benefits, justifying corrective government intervention to align private and socially optimal provision levels [34]. This makes economic capacity and public health expenditure legitimate predictors of WASH outcomes, but resource availability alone does not explain why substantial disparities persist across similarly endowed countries.

Complementing this economic foundation, global health governance frameworks emphasise that effective WASH provision requires coordinated multi-level institutional arrangements spanning international, national, and subnational actors [35]. The WHO’s health-in-all-policies approach and the WASH-specific frameworks developed under the International Drinking Water Supply and Sanitation Decade highlight that service delivery depends not only on financial resources but on regulatory coherence, accountability structures, and the capacity of governance systems to translate political commitments into operational outcomes [36, 37]. These frameworks explicitly position governance quality as a structural mediator between resource inputs and health outcomes—a perspective that welfare economics alone cannot capture. Institutional theory further reinforces this view by emphasising that formal and informal rules, norms, and accountability mechanisms shape how resources are translated into outcomes [38, 39]. In the WASH context, the capacity of regulatory bodies, corruption controls, and administrative competence condition the effectiveness of public investment, explaining why countries with similar income levels can exhibit divergent service delivery performance.

The urban environmental transition perspective sharpens our understanding of how development trajectories generate shifting WASH-specific risks. As cities grow, infrastructure investment frequently lags behind

population expansion, producing a predictable sequence of environmental health burdens [40]. In early development phases, inadequate water supply, open defecation, and poor solid waste management constitute the dominant health risks—precisely the WASH outcomes under study here. As urbanisation advances, these proximate hazards partially give way to ambient pollution and resource strain, but informal settlements and peri-urban zones continue to exhibit first-phase risk profiles even within otherwise advanced urban economics [41]. Urbanisation therefore exerts a dual and context-contingent effect on WASH outcomes: it enables economies of scale in service delivery while simultaneously generating congestion, infrastructure overloading, and spatial inequality in access, making the net effect empirically indeterminate without accounting for institutional capacity.

Political economy frameworks add a distributional dimension, highlighting that WASH improvements are rarely neutral in their allocation. Drawing on the logic of distributional conflict and analogous to Kuznets' inequality dynamics [42], growth and infrastructure investments tend to benefit better-off segments first, boosting aggregate metrics while underprivileged groups lag. Where governance is weak, elite capture and skewed political influence distort resource allocation, meaning that stronger institutions are necessary but not sufficient to guarantee equitable WASH access [43]. In synthesis, these theoretical traditions collectively suggest that WASH outcomes are not solely determined by resource endowments. Instead, they are shaped by the interaction of economic capacity, demographic dynamics, environmental risks, and governance quality. This multi-theoretical approach provides a robust foundation for analysing the complex determinants of WASH outcomes in South Asia.

Empirical background

Building on the theoretical framework, this section establishes the empirical basis for our central argument: governance quality not only directly influences WASH outcomes but also conditions the effectiveness of key structural drivers. While an extensive literature examines economic, demographic, and social determinants, their effects are empirically contingent on institutional quality. We therefore first synthesise evidence on the governance-WASH nexus, followed by a critical review of major determinants, highlighting the absence of an integrated, governance-moderated perspective. This leads to a set of novel hypotheses positioning governance as the mechanism shaping WASH outcomes in South Asia.

Governance-WASH nexus

A growing body of literature underscores the central role of governance in shaping WASH outcomes. Using cross-country data for Sub-Saharan Africa, Armah et al. [44]

show that stronger institutions improve water and sanitation coverage, even after controlling for income. Baietti et al. [45] similarly find that accountability, financial capacity, and regulatory oversight distinguish well-performing utilities, with governance failures undermining service quality and sustainability. Kayser et al. [46] demonstrate that weak coordination, enforcement, and institutional capacity constrain drinking water quality outcomes. This perspective is further reinforced by Ariffin et al. [47], who identify stakeholder engagement, policy implementation, and resource management as core pillars of effective water governance. Using global panel data, Achuo and Asongu [30] demonstrate that public spending alone does not improve WASH adoption, and may even have adverse effects in the absence of effective institutional structures. Their findings reveal that governance operates through threshold-dependent mechanisms, where the effectiveness of public investment depends on institutional quality and complementary policy conditions. Collectively, this evidence positions governance as a structural determinant of WASH outcomes, though its role in South Asia remains empirically underexplored. Therefore, it motivates the following hypothesis:

H1: Governance quality has a significant impact on improving WASH outcomes in South Asian countries.

Economic-WASH nexus

The relationship between economic capacity and WASH outcomes is complex. Some studies identify a positive association. For instance, Wang et al. [48] document long-run synergies between growth, WASH access, and infrastructure, suggesting that expanding growth resources facilitate service provision. Abdou et al. [49] demonstrate a positive correlation between WASH improvements and economic indicators in Africa. Van Minh and Hung [50] quantify the substantial global economic costs of inadequate sanitation, arguing for the high returns on WASH investments. At the micro-level, Rahut et al. [22], using data from 42 developing countries, show that higher income significantly increases access to basic WASH services. However, other evidence challenges the sufficiency of economic growth. Coffey et al. [51] demonstrate that in rural India, rising income and infrastructure expansion did not eliminate open defecation due to persistent behavioural factors. Despite these insights, the literature largely ignores how governance conditions the growth-WASH relationship, leading to the formulation of the following hypotheses:

H2: Economic growth has a significant direct effect on WASH outcomes in South Asia.

H3: Governance quality moderates the relationship between economic growth and WASH outcomes.

Demography-WASH nexus

Demographic dynamics critically shape WASH outcomes by influencing both demand and service capacity [52]. The literature underscores the dualistic nature of demographic pressures, particularly urbanisation, which can simultaneously drive development and precipitate its collapse. McDonald et al. [53] demonstrate that urban expansion outstrips municipal capacity, resulting in stark inequalities in access to piped water and sanitation. This finding is echoed at the national level; Njuguna [54], utilising logistic regression, establishes that larger household size directly increases reliance on unimproved sanitation in Kenya. The urban-rural divide is exemplified by Azeez et al. [55] in Nigeria, whose analysis confirms significantly lower access to improved water and sanitation in rural compared to urban settings. Adams et al. [56] find that the pace of demographic growth is a stronger predictor of water stress than climatic factors. Danelon et al. [57] provide a longitudinal perspective from Brazil, evidencing that internal migration from smaller to larger cities overloads sanitation systems and accelerates associated costs. While this body of work documents the direct challenges posed by demographic shifts, it largely overlooks the role of governance in mediating these effects, culminating in the development of the following hypothesis:

H4: Governance quality moderates the relationship between demographic factors and WASH outcomes.

Social-WASH nexus

Education is a key social determinant of WASH outcomes, influencing knowledge, behaviour, and adoption of improved services. The empirical literature consistently supports this relationship. Adams et al. [56], applying logistic regression to nationally representative data, show that education strongly predicts access to improved water and improved sanitation in Ghana, while Abubakar [58] confirms its independent effect in Nigeria. Further elucidating this mechanism, Emenike et al. [59] demonstrate that education elevates awareness of water quality and treatment practices in South-West Nigeria. Omole and Olajuyigbe [60] reveal that schooling levels significantly predict sanitation preferences, with more educated households exhibiting a greater propensity for investing in improved latrines and hygiene waste disposal. Corroborating this, Alam et al. [10] show that maternal education substantially improves household WASH conditions in Bangladesh, illustrating a synergistic relationship with early childhood education attainment. Despite these studies, the literature assumes a uniform effect of

education, overlooking the role of governance in shaping its impact. Therefore, we propose the following hypothesis:

H5: Education has a positive impact on WASH outcomes, with amplified effects under governance moderation.

Inequality-WASH nexus

Income inequality constitutes a structural barrier to universal WASH access by limiting affordability and skewing resource allocation. Using a 119-country panel (2004–2020), Acheampong et al. [42] show that inequality significantly lowers access to WASH, although financial inclusion mitigates this effect. At the national and sub-national levels, Kumar et al. [61] document strong pro-rich sanitation patterns in India, while Afifah et al. [62] show persistent regional disparities in Indonesia. Similar findings emerge in Peru, where Al-Kassab-Córdova et al. [63] identify significant socioeconomic inequality in WASH access and spatial clustering of disadvantage. Kumar and Kharb [11] highlight how class and caste dynamics compound sanitation and hygiene inequality in India, reinforcing that distributional politics—not just income—drive unequal WASH uptake. Yet what remains underexplored—particularly in South Asia—is how institutional quality moderates this relationship. This gap motivates the following hypothesis:

H6: Governance moderates the relationship between income inequality and WASH outcomes, reducing its negative effect.

Data and variables

We employ a panel dataset covering eight South Asian countries—Afghanistan, Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan, and Sri Lanka—over the period 2000–2024, providing the most recent and consistent data available for analysis. The focus on South Asia is motivated by both analytical and policy considerations. First, the region constitutes a relatively coherent economic and institutional setting, characterised by shared historical, cultural, and political linkages, which allows for meaningful cross-country comparison [64]. Second, despite these commonalities, South Asian countries exhibit substantial heterogeneity in governance quality, infrastructure provision, inequality, and public service delivery, providing the variations necessary to identify the key determinants of WASH outcomes [65]. Third, South Asia is one of the most densely populated and resource-constrained regions globally and faces persistent challenges related to rapid urbanisation, inequality, and climate vulnerability, all of which are closely linked to WASH outcomes [66]. These factors distinguish the region from many other parts of the world and make it

especially relevant for policy-oriented analysis. Importantly, this regional focus generates actionable insights with implications that extend beyond South Asia. By examining the region's unique interplay of institutional diversity and shared structural constraints, the analysis identifies effective governance reforms and public investment strategies that are applicable to other developing regions facing similar pressures of population growth and limited fiscal space, thereby providing a broader evidence base for both domestic and international policy design.

Dependent variables

This analysis employs three dependent variables to provide a multidimensional assessment of water, sanitation, and hygiene (WASH) infrastructures and practices. The selection of these variables is guided by their established significance in public health literature and their centrality to monitoring SDG 6. The first dependent variable, access to at least basic drinking water services (WAT), is operationalised as the percentage of the population utilising an improved water source that is located within a 30-minute round trip for collection. This metric corresponds to the basic service level, serving as a critical indicator of the population's ability to obtain safe and reliable drinking water without incurring undue time or physical burden, a fundamental determinant of health and economic productivity [67]. The second dependent variable, access to at least basic sanitation services (SAT), is defined as the percentage of the population using improved sanitation facilities that separate human excreta from human contact. This encompasses facilities such as flush/pour flush toilets connected to sewer systems or septic tanks, pit latrines with slabs, and composting toilets, excluding shared or public facilities to align with the JMP definition of basic sanitation [68]. As a core component of SDG 6.2, SAT quantifies progress in eliminating the transmission of faecal-oral diseases and ensuring dignity and safety. The third dependent variable is the prevalence of open defecation (OPR), measured as the percentage of the population that defecates in open spaces, such as fields, forests, bushes, open bodies of water, beaches, or other open areas [68, 69]. This variable functions as a direct complementary indicator to SAT, capturing the segment of the population entirely excluded from the use of any sanitation facility. The concurrent use of WAT, SAT, and OPR allows for a more nuanced examination of the sanitation landscape, particularly in the South Asian region where access to basic facilities remains inadequate.

Independent variables

To ensure robustness and mitigate omitted variable bias, this study incorporates a comprehensive set of socio-economic, environmental, and external shock indicators. These variables capture influences ranging from

endogenous structural factors to exogenous pressures, thereby providing a holistic assessment of determinants affecting the outcome variables. The explanatory variables are categorised and detailed under their respective classifications below.

Core structural

Per capita GDP (PGD, constant 2015 US\$) serves as a key indicator of economic capacity, reflecting a nation's ability to fund infrastructure, implement regulations, and elevate living standards—critical enablers for expanding water and sanitation and reducing open defecation [70]. Urbanisation (URB, % of population) captures the proportion of a country's population residing in urban areas. This variable is essential as urban settings often concentrate infrastructure investment, public service delivery, and regulatory enforcement—factors that may accelerate access to piped water, improved sanitation, and reduce open defecation [71]. However, rapid urbanisation can also strain existing systems, leading to overcrowded informal settlements with limited services. Health expenditure (HEX, % of GDP) represents the government's commitment to public health infrastructure and service delivery. Higher HEX may correlate with improved sanitation and drinking water facilities and public health campaigns [72]. School enrolment ratio (SER, % gross) reflects the level of educational participation within a country. Higher enrolment rates are associated with greater dissemination of public health knowledge, improved hygiene, and heightened community awareness [58, 73]. In addition, digitalisation (DIG) is incorporated as a structural determinant to capture the role of technological penetration and digital infrastructure in enhancing service delivery and information flows [74]. DIG is measured as the percentage of individuals using the internet. Higher levels of digitalisation are expected to improve access to water and sanitation while reducing unsafe practices such as open defecation.

Demography and inequality

Population density (POD, people per km²) measures the concentration of inhabitants per unit area. High density may strain public infrastructure, potentially overwhelming water and sanitation systems and exacerbating open defecation in underserved areas [75]. Conversely, it can also incentivise economies of scale in service provision, potentially improving access to centralised water and sanitation networks. To capture inequality, we use the GINI coefficient of disposable income (GIN), which measures post-tax and post-transfer income distribution. This reflects the effective purchasing power of households to afford and maintain access to water and sanitation services [11].

Supply pressure, shocks, and social context

Vulnerability sector score (VSS, 0–1 scale), which aggregates exposures to climate hazards, sectoral sensitivity, and adaptive capacity across water, food, health, ecosystem, infrastructure, and human habitat sectors, is used as an exogenous shock affecting water and sanitation outcomes [65]. This variable is essential to capture the systemic pressure climate vulnerability places on the dependent variables. Precipitation anomaly (PRA, z-score) is employed to quantify deviations from long-term rainfall patterns, serving as a proxy for acute climate shocks such as droughts or floods. This variable is critical for assessing how water availability extremes disrupt water supply infrastructure, compromise sanitation systems through contamination or inundation, and exacerbate open defecation by damaging facilities or limiting access to safe water sources [76].

Moreover, we innovatively construct a comprehensive governance index (GOI, on a 0–1 scale) to capture the effects of political and administrative institutional quality on water and sanitation outcomes. Incorporating governance explicitly is crucial in the South Asian context, where institutional weakness often undermines public service delivery, perpetuates inequities in resource allocation, and hinder adaptive responses to environmental or socioeconomic pressures [77, 78]. The GOI is constructed using a feedforward autoencoder neural network, an unsupervised learning technique adept at both linear and nonlinear dimensionality reduction [79, 80].

The index synthesises six governance indicators—voice and accountability, regulatory quality, political stability, corruption control, government effectiveness, and rule of law—into a single latent measure. Prior to estimation, all input variables are standardised. The network consists of a single hidden encoder layer, a one-dimensional latent bottleneck, and a symmetric decoder. Model parameters are estimated using the Adam optimiser by minimising mean squared reconstruction error. Formally, the GOI is obtained from the latent representation:

$$GOI \equiv z = \underset{\theta}{\operatorname{argmin}} \left\{ \frac{1}{n} \sum_{i=1}^n (x_i^* - \hat{x}_i)^2 \right\} \quad (1)$$

where x_i^* denotes the standardised input vector, $\hat{x}_i$ is the reconstructed output, and $\theta = (\mathcal{W}_e, \mathcal{W}_d, e, d)$ denotes the encoder and decoder parameters. The encoder compresses the six governance indicators into a lower-dimensional latent representation, which is then decoded to reconstruct the original input. To reduce overfitting and improve transparency, the model is trained using an 80/20 training-validation split, and reconstruction performance is evaluated on both samples using mean squared error. Figure 1 demonstrates the strong and stable convergence of the autoencoder, with training and validation losses declining rapidly and closely tracking each other, indicating a good model fit and no evidence of overfitting. The direction of the latent score is

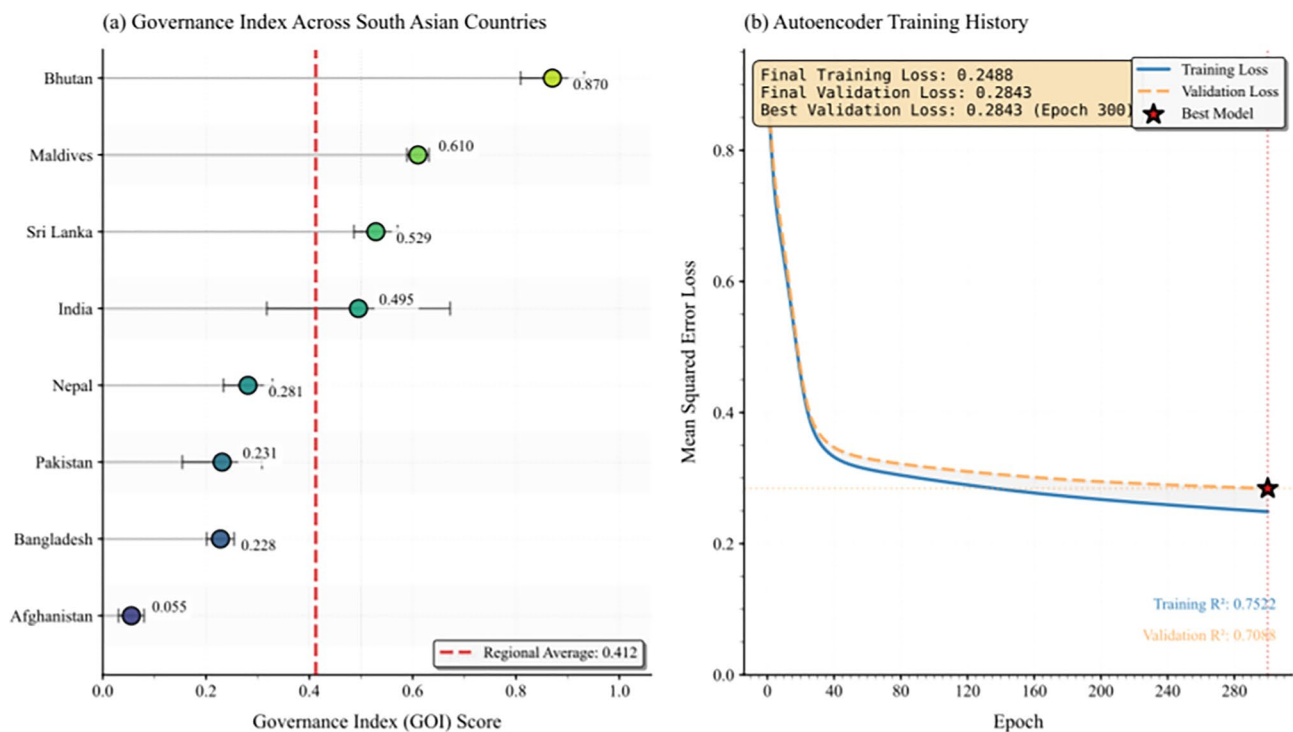


Fig. 1 GOI construction and validation

then aligned so that higher values correspond to better governance. Finally, the latent score is normalised to the [0,1] interval for interpretability. This approach avoids imposing arbitrary weights and instead allows the data to determine the relative contribution of each governance dimension, yielding a theoretically sound and empirically robust measure of governance performance [81, 82].

Finally, cultural affiliation is captured using ethnic (ETF), linguistic (LIF), and religious (REF) fractionalisation indices from Alesina et al. [83]. Each index measures the probability that two randomly selected individuals belong to different groups (ranging from 0 to 1), accounting for the role of social heterogeneity in shaping collective action and public goods provision relevant to WASH outcomes.

Sources of compilation

The dataset used in this study is constructed from a combination of internationally recognised and harmonised statistical sources to ensure reliability and comparability across countries over time. The primary dependent variables—sanitation (SAT), drinking water (WAT), and open defecation rate (OPR)—are obtained from the WHO-UNICEF Joint Monitoring Programme (JMP). Key explanatory and control variables—per capita GDP (PGD), health expenditures (HEX), population density (POD), digitalisation (DIG), and school enrolment rate (SER), as well as the indicators used to construct the governance index (GOI) are sourced from the World Bank's World Development Indicators (WDI). Finally, data for the vulnerability sector score (VSS) and precipitation anomalies (PRA) are drawn from the Notre Dame Global Adaptation Initiative (ND-GAIN), while cultural affiliation predictors are obtained from Alesina et al. [83].

Estimation methods

Empirical models

Our empirical approach is grounded in urban environmental transition and institutional economic theories (see Sect. “Literature review”), which posit that governance quality is a primary mechanism translating economic and demographic factors into tangible public goods like WASH services. The baseline specification models of WASH outcomes Y_{it} —sanitation (SAT), water access (WAT), or open defecation rate (OPR)—as a function of key structural, economic, and environmental determinants identified in the extant literature:

$$Y_{it} = \theta_i + \sum_{k=1}^K \beta_k X_{it} + \varepsilon_{it} \quad (2)$$

where θ_i captures the unobserved time-invariant country-level heterogeneity, β_k represents the vector

of coefficients for the explanatory variables (X_{it}), and ε_{it} denotes the idiosyncratic error. To test our central hypothesis—that GOI operates as an effect modifier rather than a direct driver—we extend Equation [2] to include interaction terms between GOI and all covariates:

$$Y_{it} = \theta_i + \sum_{k=1}^8 \beta_k X_{kit} + \lambda GOI_{it} + \sum_{k=1}^8 \vartheta_k (X_{kit} \times GOI_{it}) + \varepsilon_{it} \quad (3)$$

where all vectors retain similar definitions, and ϑ_k captures the moderating effects of GOI on the relationships between Y_{it} and X_{it} . With this, we empirically shift focus from interpreting main effects in isolation to estimating the conditional effects calculated from both the main and interaction coefficients.

Estimation strategy

Classical panel estimators rely on a set of assumptions—including cross-sectional independence, parameter homogeneity, and error stationarity—that provide a tractable and widely used framework for empirical analysis [84, 85]. When these conditions are reasonably satisfied, such methods offer efficient, interpretable, and well-established tools that have been successfully applied across a wide range of economic contexts [86]. However, in complex, interdependent socio-economic settings such as South Asia, these assumptions are often empirically violated, as confirmed by our diagnostic tests (see Tables 1 and 2). In practice, imposing such assumptions treats distinct national contexts as statistically exchangeable, effectively imposing a single, uniform effect for each covariate across highly heterogeneous institutional, economic, and geographic environments [87]. This not only obscures country-specific structural dynamics but also introduces aggregation bias, producing “one-size-fits-all” coefficient estimates that may be misleading or lack substantive interpretability [86]. Furthermore, while the frequentist econometric paradigm provides a rigorous foundation for statistical inference, it faces limitations in explicitly addressing model and parameter uncertainty [88]. Conventional methods rely on large-sample asymptotics for inference [89], which may be less reliable in finite and unbalanced panels typical of regional studies. Even advanced corrections—such as Driscoll and Kraay [90] standard errors or common correlated effects estimators [91]—partially mitigate these issues but remain constrained within an asymptotic framework, which can yield unstable inference in the presence of strong cross-sectional spillovers.

In response to these challenges, we adopt the Bayesian Hierarchical Linear Model (HLM), originally proposed by Bryk and Raudenbush [92], as our core estimator. Rather than replacing classical approaches,

Table 1 Summary statistics and correlation

(a) Summary statistics		SAT	WAT	OPR	PGD	URB	HEX	SER	POD	GIN	VSS	PRA	GOI	ETF	LIF	REF	DIG
Mean		39.52	49.85	16.91	2436.47	28.65	34.03	101.90	466.29	38.17	0.53	0.96	0.41	0.421	0.466	0.235	17.495
SD		30.45	20.96	17.87	2666.15	7.92	21.45	18.83	471.18	5.13	0.03	199.96	0.25	0.270	0.315	0.136	20.957
Min.		5.65	18.24	0.00	277.11	13.39	3.30	22.16	14.997	31.40	0.45	-715.99	0.00	0.003	0.000	0.000	0.000
Max.		98.59	99.70	73.33	11194.92	43.68	79.99	150.35	1747.02	48.60	0.61	730.94	1.00	0.045	0.092	0.154	2.468
(b) Correlation matrix		SAT	WAT	OPR	PGD	URB	HEX	SER	POD	GIN	VSS	PRA	GOI	ETF	LIF	REF	DIG
SAT		1.000	0.010	-0.539	-0.064	-0.264	0.432	-0.187	0.399	0.324	-0.271	0.094	0.245	-0.304	-0.609	-0.065	0.221
WAT			1.000	0.158	0.910	0.169	0.118	0.519	0.005	-0.117	-0.086	0.060	0.041	0.065	0.117	-0.477	-0.203
OPR				1.000	0.175	-0.079	-0.490	0.152	-0.300	0.105	0.151	-0.058	-0.261	0.376	0.376	0.644	-0.006
PGD					1.000	0.377	0.050	0.644	-0.049	-0.058	-0.206	0.140	0.030	0.116	0.207	-0.362	-0.009
URB						1.000	0.087	0.507	-0.289	0.381	-0.506	0.117	0.238	0.084	0.130	0.544	0.332
HEX							1.000	-0.192	0.133	0.374	-0.350	0.020	0.802	-0.394	-0.326	-0.037	0.172
SER								1.000	0.012	0.074	-0.053	0.100	0.044	-0.047	0.065	-0.111	0.036
POD									1.000	-0.108	0.177	-0.045	0.000	-0.869	-0.749	-0.481	0.080
GIN										1.000	-0.650	0.117	0.552	-0.105	-0.062	0.603	0.340
VSS											1.000	-0.174	-0.285	-0.115	-0.109	-0.553	-0.515
PRA												1.000	0.030	0.051	0.018	0.067	0.102
GOI													1.000	-0.396	-0.203	0.056	0.160
ETF														1.000	0.797	0.335	-0.029
LIF															1.000	0.134	0.066
REF																1.000	0.266
DIG																	1.000

Notes: SD: Standard deviations, Min.: Minimum, Max.: Maximum, SAT: Sanitation access, WAT: Water access, OPR: Open defecation, PGD: Per capita GDP, URB: Urbanisation, HEX: Health expenditures, SER: School enrolment ratio, POD: Population density, GIN: Gini coefficient, VSS: Vulnerability sector score, PRA: Precipitation anomalies, GOI: Governance quality index, ETF: Ethnic fractionalisation, LIF: Linguistic fractionalisation, REF: Religious fractionalisation, DIG: Digitalisation

Table 2 Cross-sectional dependence and slope heterogeneity

Tests estimated	Model I: SAT		Model II: WAT		Model III: OPR	
	Statistics	p-value	Statistics	p-value	Statistics	p-value
<i>Cross-sectional dependence tests</i>						
Breusch–Pagan LM test	282.245***	0.000	143.718***	0.000	195.001***	0.000
Pesaran CD test	6.440***	0.000	−0.931	0.350	2.331**	0.020
<i>Slope heterogeneity test</i>						
Delta	11.201***	0.000	9.111***	0.000	6.940***	0.000
Adjusted delta	9.066***	0.000	9.236***	0.000	8.072***	0.000
<i>Heteroskedasticity tests</i>						
Breusch–Pagan/Cook–Weisberg test	0.110	0.625	1.044	0.345	0.266	0.505
Wooldridge test for autocorrelation	2044.15***	0.000	1076.12***	0.000	940.42***	0.000

***denotes significance at 1% level

this framework complements them by relaxing restrictive assumptions and providing a more flexible structure for modelling heterogeneity and uncertainty. The HLM incorporates country-level heterogeneity through partially pooled intercepts [93], which borrow strength across units to stabilise estimates while preserving contextual nuance—striking a balance between the excessive noise of fixed effects and biased pooling of aggregate models [94]. Crucially, Bayesian inference provides complete posterior distributions for all parameters, enabling probabilistic statements about effect sizes (e.g., $\mathbb{P}(\beta > 0)$) rather than reliance on asymptotic *p-values* [95]. This is especially valuable in small- to moderate-sized samples, where asymptotic properties may not hold. By integrating hierarchical structure with Bayesian updating, our model simultaneously accounts for unobserved heterogeneity, cross-sectional dependence, and temporal persistence within a coherent probabilistic structure [96]. The HLM is formally specified as:

$$\begin{aligned}
 Y_{it} &\sim \mathcal{N}(\mu_{it}, \sigma^2), \\
 \mu_{it} &= \theta_i + X_{it}\beta, \\
 \theta_i &\sim \mathcal{N}(\mu_{it}, \sigma_\theta^2),
 \end{aligned} \tag{4}$$

where Y_{it} denotes the WASH outcome (WAT, SAT, or OPR) for country i at time t , assumed to follow a normal distribution with mean μ_{it} and variance σ^2 . The conditional mean μ_{it} consists of a country-specific intercept θ_i , which captures unobserved time-invariant heterogeneity across countries, and a linear component $X_{it}\beta$, representing the contribution of observed covariates. The intercept θ_i is modelled hierarchically as a random draw from a common distribution with mean μ_0 and variance σ_θ^2 , allowing partial pooling across countries. This hierarchical structure enables the model to borrow information across units while preserving country-specific variation. We assign weak informative priors $\beta_k \sim \mathcal{N}(0, 1)$, $\theta_i \sim \mathcal{N}(0, 10)$, $\sigma, \sigma_\theta \sim t_3(0, 10)$ to ensure computational

stability without imposing strong prior beliefs [97]. This specification delivers inferences that are both empirically robust and theoretically grounded in the contextual realities of South Asia. To estimate the conditional effects of covariates moderated by GOI, we augment the HLM model with interaction terms:

$$\begin{aligned}
 Y_{it} &\sim \mathcal{N}(\mu_{it}, \sigma^2), \\
 \mu_{it} &= \theta_i + \sum_{k=1}^8 \beta_k X_{kit} + \lambda GOI_{it} + \sum_{k=1}^8 \vartheta_k (X_{kit} \times GOI_{it}) + \varepsilon_{it}, \\
 \theta_i &\sim \mathcal{N}(\mu_{it}, \sigma_\theta^2), \beta_k, \lambda, \vartheta_k \sim \mathcal{N}(0, 10).
 \end{aligned} \tag{5}$$

where all vectors hold similar meanings as defined before. In this framework, the marginal effect of covariates X_k is given by $\beta_k + \delta_k \times GOI$. A significant ϑ_k indicates that governance quality moderates the effect of X_{it} on the WASH outcomes. However, the effects of X_{it} remain constant and do not vary with governance when ϑ_k is not significantly different from zero.

Robustness checks

To ensure the credibility of our inferences, we implement a three-pronged robustness strategy. First, we conduct a prior sensitivity analysis by re-estimating the model under a more diffuse normal prior $\mathcal{N}(0, 5)$, which relaxes shrinkage towards zero relative to our baseline weakly informative $\mathcal{N}(0, 1)$ specification. Second, we compare our HLM specification against a pooled Bayesian generalised linear model (GLM) as a counterfactual alternative:

$$\begin{aligned}
 Y_{it} &\sim \mathcal{N}(\mu_{it}, \sigma^2), \\
 \mu_{it} &= \alpha + \sum_{k=1}^8 \beta_k X_{kit} + \varepsilon_{it},
 \end{aligned} \tag{6}$$

$$\mu_{it} = \alpha + \sum_{k=1}^8 \beta_k X_{kit} + \lambda GOI_{it} + \sum_{k=1}^8 \vartheta_k (X_{kit} \times GOI_{it}) + \varepsilon_{it},$$

$$a \sim \mathcal{N}(\mu_{it}, \sigma_a^2), \beta_k, \lambda, \vartheta_k \sim \mathcal{N}(0, 10),$$

If the GLM—which assumes parameter homogeneity across countries—produces markedly different or less precise estimates, it provides empirical evidence that accounting for between-country heterogeneity is not merely a statistical refinement but an empirical necessity for credible inference in this setting [98, 99]. Third, to address potential reverse causality between WASH outcomes and economic development, we estimate a Bayesian structural control function specification treating PGD as a potentially endogenous regressor. Following the control function approach of Wooldridge [100], we instrument PGD using its own two-period lagged values (PGD_{t-1} , PGD_{t-2}), which satisfy the exclusion restriction under the assumption that past income levels affect current WASH outcomes only through their effect on current income. The two-stage specification is:

First stage

$$PGD_{it} = \alpha_0 + \alpha_1 PGD_{i,t-1} + \alpha_2 PGD_{i,t-2} + \sum_k \gamma_k X_{kit} + \xi_{it}, \quad (7)$$

Second stage

$$Y_{it} = u_i + \beta_{PGD} \cdot PGD_{it} + \sum_k \beta_k X_{kit} + \lambda \hat{\xi}_{it} + \varepsilon_{it}. \quad (8)$$

where u_i represents country-specific random intercepts estimated under a non-centred parameterisation, and $\hat{\xi}_{it}$ is the first-stage residual whose coefficient λ directly tests for endogeneity. A posterior distribution of λ that is not credibly different from zero indicates the absence of simultaneity bias. All priors are weakly informative: $\beta_k \sim \mathcal{N}(0, 5)$, $\lambda \sim \mathcal{N}(0, 1)$, $\sigma_{ID} \sim \text{HalfNormal}(20)$.

We estimate the model using 4,000 warmup and 4,000 sampling iterations per chain across four chains, with an acceptable target acceptance rate of 0.99 to ensure reliable exploration of the posterior geometry.

Statistical results

Descriptive statistics

Table 1 presents the summary statistics, revealing considerable heterogeneity across the region. It indicates that, on average, access to basic services remains a critical challenge: merely 49.85% of the population enjoys at least basic drinking water (WAT) and 39.52% have access to basic sanitation (SAT). Conversely, open defecation (OPR) persists among 16.91% of the population. Economic capacity, proxied by per capita GDP

(PGD), exhibits extreme heterogeneity, with a mean of \$2,436 but ranging from \$277–\$11,194. This reflects the substantial development divide between the low- and highest-income countries in the sample. The region remains predominantly rural, with an average urbanisation rate (URB) of 28.65% (on average), which poses distinct challenges for infrastructure and service delivery. Commitment to public health, as measured by health expenditure (HEX), averages 34% of GDP. School enrolment rate (SER) approaches 101.9% in gross terms, indicating broad access to basic education, though potentially inflated by over-age enrolment. Population density (POD) varies strikingly, from sparse to extremely dense settlements (max = 1747 people/km²), with implications for the cost and modality of service provision. Income inequality, captured by the Gini coefficient (GIN), averages 38.17, indicating a moderate level of disparity. The vulnerability sector score (VSS) averages 0.53, signalling systemic exposure to climatic and environmental risks. Precipitation anomalies (PRA) show extreme variability, highlighting the region's exposure to variable and unpredictable climate shocks. Finally, governance quality (GOI) spans the full range (0–1), emphasising the profound institutional divergence across South Asia.

Cultural affiliation variables show considerable heterogeneity: ETF averages 0.421, LIF averages 0.466, and REF averages 0.235. Digitalisation (DIG) has a mean value of 17.5%, with substantial variations (0–2.5%), reflecting uneven technological penetration across the region. The correlation matrix is examined to assess potential multicollinearity among explanatory variables. It is important to emphasise that these correlations reflect co-movement between variables and do not imply causal relationships. Several pairwise correlations are nonetheless theoretically informative. A moderate positive association exists between HEX and GOI ($r=0.808$), consistent with the hypothesis that effective governance facilitates greater public health investment. A moderate negative correlation is observed between URB and VSS ($r=-0.578$), indicating that urban contexts may be associated with lower systemic vulnerability. LIF is strongly negatively correlated with SAT ($r=-0.609$), while REF is positively correlated with OPR ($r=0.644$), suggesting that social heterogeneity may impede collective action around sanitation outcomes. The remaining correlations are generally weak, suggesting that the model's independent variables capture distinct structural, demographic, and shock-related dimensions, thereby mitigating concerns of multicollinearity.

Table 2 reports diagnostic tests for cross-sectional dependence, slope heterogeneity, heteroskedasticity, and serial correlation across the three empirical models (Model I: SAT; Model II: WAT; Model III: OPR). The Breusch-Pagan [101] LM test strongly rejects the null of

cross-sectional independence at the 1% level, indicating substantial residual dependence and the presence of unobserved common factors or regional spillovers. Conversely, the Pesaran [102] test yields statistically insignificant results. This divergence reflects known differences in test properties: the LM test is more powerful in panels with relatively small cross-sectional dimension (N) and larger time dimension (T), whereas the CD test is designed for large-N panels and may lose power in smaller samples. Given the sample structure of this study, greater weight is therefore placed on the LM test results. The Pesaran and Yamagata [103] test rejects the null of slope homogeneity at the 1% level across all models, confirming substantial cross-country heterogeneity in economic structures, institutional capacities, and developmental pathways in the region. The Breusch–Pagan/Cook–Weisberg test indicates the absence of heteroskedastic disturbances across all specifications. Finally, Wooldridge’s test strongly rejects the null of no serial correlation, indicating that shock persistence is a salient feature of the data-generating process.

Direct effects

The results of the Bayesian HLM reported in Table 3 provide a complete probabilistic assessment of the direct relationships between the explanatory variables and the three WASH outcomes—sanitation access (SAT), water access (WAT), and open defecation (OPR). The results

reveal a selective pattern of effects, with only a subset of variables showing credible posterior distributions, while many others remain centred around zero. This suggests that WASH outcomes are shaped by a limited number of dominant structural drivers rather than by all included covariates equally.

Among the core structural factors, per capita GDP (PGD) shows no credible direct effect on any of the three outcomes, suggesting that aggregate national wealth alone is insufficient for improving WASH access without targeted policies. This finding reflects the nature of WASH as a non-market, infrastructure-intensive public good, where improvements depend on coordinated, sector-specific investment rather than broad income expansion. Aggregate income may also mask substantial inter-country disparities, with economic gains concentrated in urban or higher-income segments while underserved populations remain excluded. As a result, increases in national income do not automatically translate into expanded or equitable access to WASH services. By contrast, urbanisation (URB) emerges as one of the strongest and most consistent predictors. It is positively associated with sanitation access, but negatively linked with water access and open defecation. Health expenditures (HEX) also demonstrate robust effects in improving sanitation and reducing open defecation. Similarly, the school enrolment ratio (SER) is positively associated with water access and negatively with open defecation,

Table 3 Bayesian HLM estimates of direct effects

Parameters	Model-I: SAT			Model-II: WAT			Model-III: OPR		
	Coeff	CrI lower	CrI upper	Coeff	CrI lower	CrI upper	Coeff	CrI lower	CrI upper
Intercept	−45.829	−88.42	−6.00	−45.164	−87.92	−3.09	−44.446	−94.54	4.62
PGD	0.001	0.00	0.00	0.001	0.00	0.00	−0.002	0.00	0.00
URB	1.331	1.05	1.62	−0.710	−1.08	−0.33	−0.925	−1.47	−0.43
HEX	0.153	0.06	0.26	0.077	−0.05	0.21	−0.283	−0.46	−0.10
SER	0.027	−0.03	0.08	0.200	0.13	0.28	−0.139	−0.24	−0.04
POD	0.009	0.00	0.02	0.036	0.03	0.05	0.017	0.00	0.03
GIN	0.954	0.37	1.52	1.879	1.14	2.63	2.610	1.64	3.57
VSS	−9.684	−28.75	8.41	−4.618	−23.02	14.50	12.149	−6.84	31.82
PRA	0.002	0.00	0.00	0.001	0.00	0.00	−0.004	−0.01	0.00
GOI	−3.422	−10.00	3.45	−1.750	−9.99	7.00	−0.296	−11.12	10.15
ETF	−0.443	−19.44	18.70	3.920	−16.10	22.41	3.848	−16.13	22.02
LIF	−2.548	−22.23	15.98	4.152	−14.85	23.35	6.758	−12.21	26.43
REF	−0.816	−19.89	19.76	−0.456	−20.00	18.23	−0.515	−19.27	18.83
DIG	0.093	0.05	0.13	−0.046	−0.10	0.00	−0.167	−0.24	−0.10
Model diagnostics									
SD (intercept)	40.805	23.40	74.41	32.765	18.02	59.84	26.810	13.70	49.62
σ (residual)	3.204	2.83	3.55	4.251	3.78	4.72	5.820	5.19	6.49
R-hat (all params)	1.000			1.000			1.000		
Bulk ESS (average)	5781			7596			5517		
Number of chains	4 (8)			4 (8)			4		
Number of iterations	4000			4000			4000		

Note: Entries are posterior means with 95% credible intervals (CrI). s is the residual standard deviation; SD (intercept) reflects cross-country variation in intercepts. R-hat indicates satisfactory MCMC convergence; Bulk ESS reports effective sample size

highlighting education's role in promoting public health knowledge and hygiene practices.

With respect to demographic and distributive factors, population density (POD) shows a robust positive association with all three outcomes. The positive effects on SAT and WAT suggest that denser populations may benefit from economies of scale in infrastructure provision. At the same time, the positive coefficient for OPR implies that demographic pressure may also intensify strain on sanitation systems in some contexts, so that concentration generates both service expansion and deprivation. The Gini coefficient (GIN) reveals a counterintuitive yet strong positive association with all outcomes, particularly in reducing open defecation. Higher measured inequality may reflect ongoing structural transformation, where initial improvements in WASH services are concentrated in urban or higher-income areas. This pattern would generate aggregate gains in coverage while simultaneously widening measured inequality, as progress is driven by targeted investments or spatially uneven development rather than broad-based inclusion. Alternatively, this result could stem from measurement or aggregate effects, where national-level inequality indices do not fully capture localised disparities in WASH access. Accordingly, these findings warrant cautious interpretation and should be viewed as indicative of uneven development patterns rather than evidence of inclusive progress.

The vulnerability sector score (VSS) and precipitation anomalies (PRA) both have posterior intervals that include zero in all models, indicating that short-run exposure and climatic disturbances do not exert stable direct effects once broader structural conditions are controlled for. Likewise, the governance quality index (GOI) shows no direct effect on any WASH outcomes, suggesting that institutional quality may operate through indirect or moderating pathways. Cultural affiliation proxies (ETF, LIF, and REF) exhibit wide credible intervals crossing zero across all specifications, indicating no credible direct associations with WASH outcomes. By contrast, digitalisation (DIG) emerges as a significant predictor. It credibly enhances SAT, while largely reducing OPR in the region. Moreover, the model diagnostics indicate excellent estimation properties. All R-hat values are one, indicating Markov Chain Monte Carlo (MCMC) convergence, while effective sample sizes are sufficiently large to ensure posterior reliability. The substantial country-level intercept variation confirms significant unobserved heterogeneity between nations, validating the hierarchical modelling approach.

Figures 1A–3A of Appendix A present prior-posterior densities and MCMC diagnostics for the three models. Across all specifications, posterior distributions are more concentrated than diffuse priors, confirming strong data informativeness. The most pronounced posteriors

are observed for URB, HEX, GIN, POD, DIG, and SER, reflecting their substantive influence on WASH outcomes. In contrast, variables such as GOI, VSS, PRA, ETF, LIF, and REF exhibit posteriors centred close to zero, consistent with the absence of credible direct effects. The MCMC trace plots show well-mixed and stationary chains across all parameters, with no evidence of non-convergence.

Moderating effects

Table 4 presents the Bayesian HLM model estimates investigating the moderating role of GOI on the relationship between structural determinants and WASH outcomes. Consistent with the direct effects (Table 3), GOI does not exhibit a statistically credible standalone impact. However, the interaction models demonstrate that GOI operates as an important conditional factor, altering both the magnitude and direction of several key relationships.

The moderating role of GOI is selective rather than universal, with statistically credible effects concentrated in a subset of variables. For PGD, the interaction terms are small but positive for sanitation and water access. However, the interaction for open defecation is not statistically distinguishable from zero, indicating no credible moderating role in this dimension. This indicates that good governance is essential for translating aggregate national wealth into tangible WASH improvements. GOI significantly amplifies the positive effect of URB on sanitation and increases its association with open defecation. Meanwhile, the interaction effect for water access is not credible. This indicates that while governance enhances the infrastructure and behavioural consequences of urbanisation in some dimensions, it does not uniformly resolve pressures on water systems. Importantly, the positive interaction with OPR suggests that governance may be associated with uneven urban transition, where improvements in some areas coexist with persistent deprivation in others. For health expenditures (HEX), governance significantly reduces its effect on sanitation and water access, while increasing its association with open defecation. This indicates that the effectiveness of health spending is conditional on governance structures, and that higher governance quality does not uniformly enhance the translation of health investments into WASH improvements. Instead, it may reflect differences in allocation efficiency or competing policy priorities across sectors. A similar pattern is observed for the school enrolment ratio (SER). Governance weakens its effect on sanitation and water access, while the interaction with open defecation is not statistically credible. This suggests that while education remains important, its effectiveness in shaping WASH outcomes may depend less on governance quality than expected or may operate

Table 4 Bayesian HLM estimates—moderating role of GOI

Parameter	Model-I: SAT			Model-II: WAT			Model-III: OPR		
	Coeff	CrI Lower	CrI Upper	Coeff	CrI Lower	CrI Upper	Coeff	CrI Lower	CrI Upper
PGD × GOI	0.005	0.00	0.01	0.006	0.00	0.01	−0.005	−0.01	0.00
URB × GOI	1.060	0.12	2.05	−0.250	−1.40	0.85	2.393	1.05	3.66
HEX × GOI	−0.380	−0.64	−0.09	−0.430	−0.76	−0.10	0.511	0.12	0.94
SER × GOI	−0.303	−0.52	−0.07	−0.347	−0.63	−0.08	0.209	−0.12	0.53
POD × GOI	−0.001	−0.03	0.02	0.023	0.00	0.05	0.022	−0.01	0.06
GIN × GOI	0.399	−0.24	1.06	0.960	0.14	1.71	−2.726	−3.66	−1.75
VSS × GOI	0.016	−1.95	1.97	0.015	−2.03	1.93	−0.030	−1.95	2.02
PRA × GOI	−0.004	−0.01	0.00	−0.001	−0.01	0.01	0.011	0.00	0.03
ETF × GOI	0.060	−1.86	2.01	0.105	−1.76	2.08	−0.101	−2.12	1.72
LIF × GOI	0.022	−2.00	2.05	0.068	−1.91	2.02	−0.091	−1.99	1.88
REF × GOI	0.018	−2.00	1.91	0.088	−1.91	2.09	−0.022	−1.91	1.92
DIG × GOI	−0.462	−0.70	−0.22	−0.627	−0.91	−0.32	0.137	−0.20	0.49
Model diagnostics									
SD (intercept)	44.168	23.95	80.50	37.319	18.76	72.28	32.894	18.52	57.02
σ (residual)	2.918	2.59	3.25	3.647	3.24	4.07	4.614	4.08	5.17
R-hat (all params)	1.000			1.000			1.000		
Bulk ESS (average)	7665			5838			7617		
No of chains	4			4			4		
No of iterations	4000			4000			4000		

Note: Entries are posterior means with 95% credible intervals (CrI). s is the residual standard deviation; SD (intercept) reflects cross-country variation in intercepts. R-hat indicates satisfactory MCMC convergence; Bulk ESS reports effective sample size

through indirect behavioural channels not fully captured in the model.

Population density (POD) shows no credible interaction effect across all models, indicating that its influence operates largely independently of governance conditions. Likewise, the vulnerability sector score (VSS) and precipitation anomalies (PRA) do not exhibit statistically significant moderation effects, reinforcing the conclusion that short-term shocks and structural vulnerability factors are not systematically conditioned by governance in this framework. In contrast, income inequality (GIN) demonstrates one of the strongest and most consistent moderating patterns. Governance significantly amplifies the positive association between inequality and sanitation and water access, while strongly reducing open defecation. This suggests that governance quality plays a critical role in shaping how unequal resource distributions translate into aggregate outcomes, potentially reflecting targeted investments or spatially uneven development processes. Interestingly, the interaction of GOI with digitalisation (DIG) is negative and statistically significant for both SAT and WAT, indicating that the positive contribution of digitalisation diminishes as governance quality improves. This strongly reflects diminishing returns and a substitution effect, where digital tools play a stronger role in a moderate governance environment but become less pivotal as institutional quality jumps over a certain threshold. For OPR, the interaction term of GOI with DIG is credibly insignificant, implying that the behavioural impact of digitalisation is not systematically

conditioned by governance. The remaining cultural variables (ETF, LIF, and REF) show no credible interaction across any outcomes, highlighting that their influence is neither direct nor conditionally shaped by governance within this framework.

The model diagnostics indicate reliable estimation, with R-hat at 1.000 indicating successful MCMC convergence, and effective sample sizes exceeding 4,000 draws ensuring posterior reliability. The substantial country-level intercept variation across all models confirms significant unobserved heterogeneity between nations. Figures 4A–6A of Appendix A display prior-posterior densities and MCMC trace plots for the interaction models. Posterior distributions are narrower than priors, confirming that the data strongly inform the interaction effects, with particularly notable signals for the role of governance quality in enhancing the relationships between the explanatory variables and WASH outcomes. Trace plots show stable mixing across chains with no evidence of convergence issues, supporting the reliability of the estimates. To further characterise the conditional nature of the interaction effects, we employ a Johnson-Neyman (JN)-type interpretation within our Bayesian framework to identify the ranges of GOI over which the marginal effects of key variables become statistically credible. Rather than focusing on a single point estimate, this approach allows the marginal effect of each variable to vary continuously across the observed distribution of GOI, thereby providing a more nuanced understanding of the moderation mechanism.

Figure 2 illustrates how GOI conditions the relationships between key determinants and sanitation access. The marginal effect of PGD becomes positive and statistically significant only at higher levels of governance, indicating that income gains translate into sanitation improvements primarily in well-governed contexts. URB maintains a consistently positive effect across all the governance spectrum, though with limited statistical

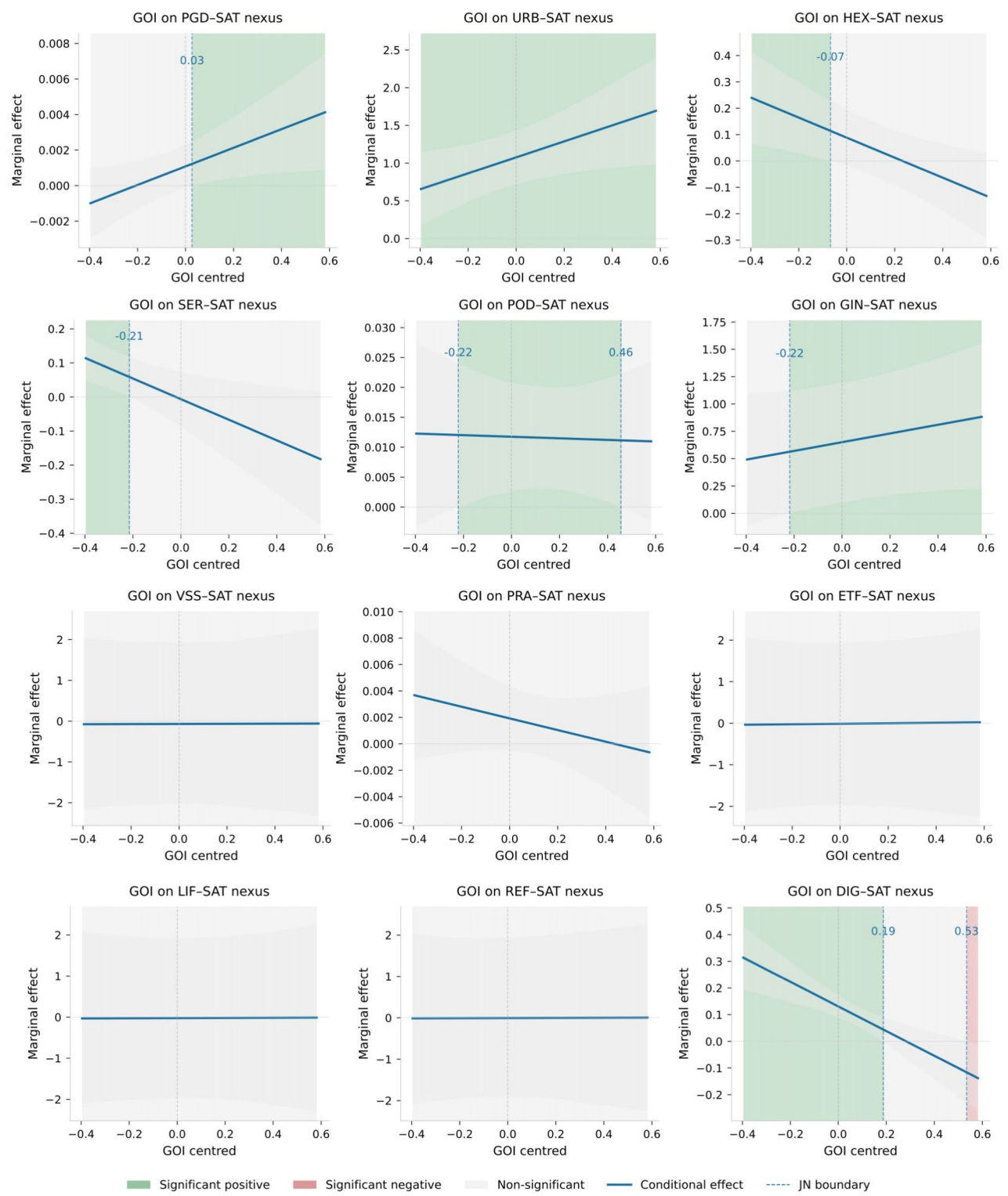


Fig. 2 Marginal effect plots for Model-I: SAT

precision, suggesting that its benefits are broadly robust but not governance-dependent. In contrast, HEX and SER exhibit negative and significant slopes as governance improves, implying diminishing marginal returns at higher governance levels. GIN shows a strongly positive and increasing effect with better governance, reinforcing the role of institutional quality in transforming unequal development into sanitation gains. VSS, PRA, ETF, LIF, and REF remain largely insignificant across the GOI range, indicating governance-independent effects. However, the impact of digitalisation remains significantly positive in a lower GOI range (below 0.19), becomes insignificant between 0.20–0.53, and then turns negative afterwards.

Figure 3 presents the governance-conditioned marginal effects for water access, revealing a markedly different pattern. PGD becomes increasingly positive and significant as GOI improves, highlighting that institutional quality is essential for converting income gains into water infrastructure expansion. In contrast, URB exhibits a strongly negative effect at higher levels of governance, suggesting that rapid urban growth may strain water systems even under better institutional conditions. GIN shows a robust and positive marginal effect, indicating that improvements in water access may initially be concentrated in more advantaged segments. HEX and SER display weak moderating effects, implying limited governance conditioning in these channels. While POD remains positive under GOI moderation, VSS, PRA, ETF, LIF, and REF exhibit clear governance-independent patterns in the region. Among others, DIG indicates a positive interaction in weaker institutional environments but becomes significantly negative once GOI improves beyond 0.05.

Figure 4 depicts the marginal effects of GOI's moderating role on open defecation. PGD shows a consistent negative effect on open defecation as GOI improves. Conversely, under GOI moderation, the marginal positive impact of URB on open defecation weakens beyond 0.24, implying that—conditional on governance—urban expansion is initially associated with higher open defecation, but this effect is neutralised by governance beyond a certain threshold. A similar pattern is observed for HEX. POD is positively associated with open defecation; however, as GOI improves, this effect increases substantially. GIN has a negative marginal effect on OPR across the entire range of GOI. PRA remains positive and statistically significant in amplifying its own effect; yet beyond a certain GOI threshold (i.e., 0.17), its impact is neutralised. SER, VSS, ETF, LIF, and REF remain credibly insignificant. Finally, the marginal effect of DIG is positive on OPR; however, GOI offsets this positivity beyond a threshold value of 0.36.

Robustness test results

To evaluate the stability and reliability of our empirical findings, we implemented three robustness checks, the results of which are reported in Tables 5, 6 and 7. First, the baseline Bayesian HLM models were re-estimated using a more diffuse prior distribution, replacing the weakly informative *Normal* (0, 1) priors with wider *Normal* (0, 5) priors. The results demonstrate a high degree of robustness across specifications. The direction, statistical credibility, and substantive magnitude of the coefficients remain largely unchanged under the wider priors. As theoretically expected, minor shifts in posterior point estimates emerge under the more diffuse specification, yet the extensive overlap in the posterior credible intervals across all three outcomes indicates that the core inferential conclusions are not sensitive to prior choice. This consistency provides strong evidence that the reported effects are not artefacts of prior assumptions. Second, the Bayesian HLM estimates were compared with those obtained from a pooled Bayesian GLM framework that excludes country-level random intercepts. This comparison evaluates whether the hierarchical structure—capturing unobserved country-specific heterogeneity—drives the results, or whether similar inferences emerge from a simpler pooled specification. Overall, the findings remain broadly consistent across the two specifications. Key differences such as GIN, POD, and DIG retain similar signs and, in most cases, overlapping credible intervals across models, as further illustrated in the forest plot of comparative analysis (Fig. 5).

However, several differences emerge. Notably, URB in the SAT equation exhibits a sign reversal, switching from a positive credible effect in the HLM model to a negative and statistically credible effect in the pooled GLM. This divergence is consistent with the interpretation that unobserved country-level factors—absorbed by the random intercepts in the HLM—confound the pooled GLM estimates. Similarly, HEX shows a stronger positive effect in the GLM for SAT and WAT, suggesting that pooled models may overstate its impact when country-specific heterogeneity is not accounted for. For several variables, including SER and GOI, the GLM estimates become statistically insignificant or unstable, further reinforcing the importance of the hierarchical specification. Third, to address potential reverse causality between WASH outcomes and economic development, we estimated a Bayesian structural control function specification treating PGD as a potentially endogenous regressor, using two-period lagged values of PGD as instruments. The posterior distributions of the endogeneity parameter (λ) were not credibly different from zero in any of the three outcome equations, with credible intervals comfortably spanning zero in all cases. This indicates no statistical evidence of simultaneity bias, and the baseline hierarchical estimates

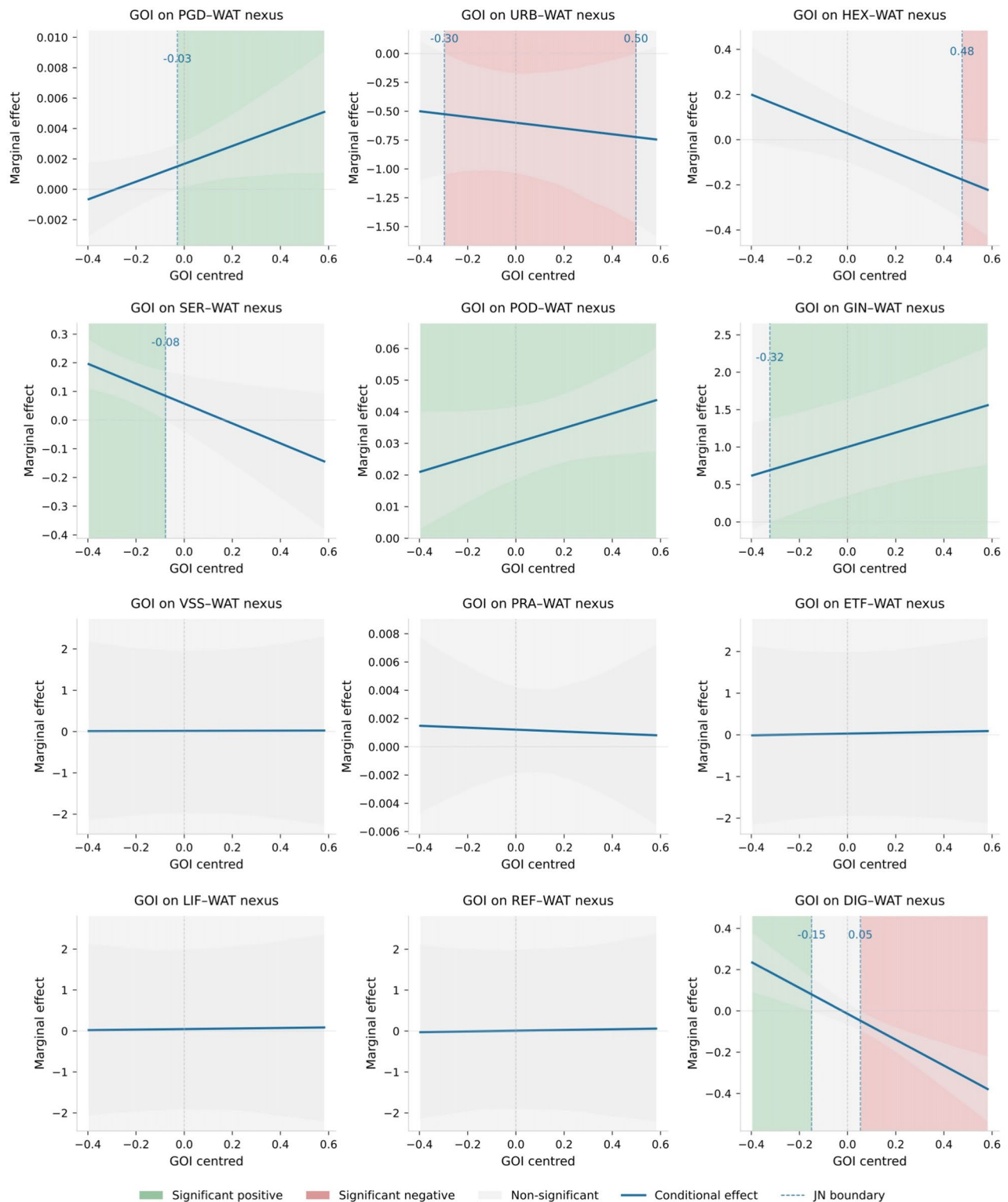


Fig. 3 Marginal effect plots for Model-II: WAT

are consequently not materially affected by endogeneity. Convergence diagnostics confirm reliable estimation across all three specifications, with R-hat values of 1.000 for all parameters and adequate effective sample sizes

throughout. Taken together, these checks demonstrate that our results are not artefacts of prior assumptions model structure, or endogenous regressors. Instead, they reflect robust and stable empirical relationships between

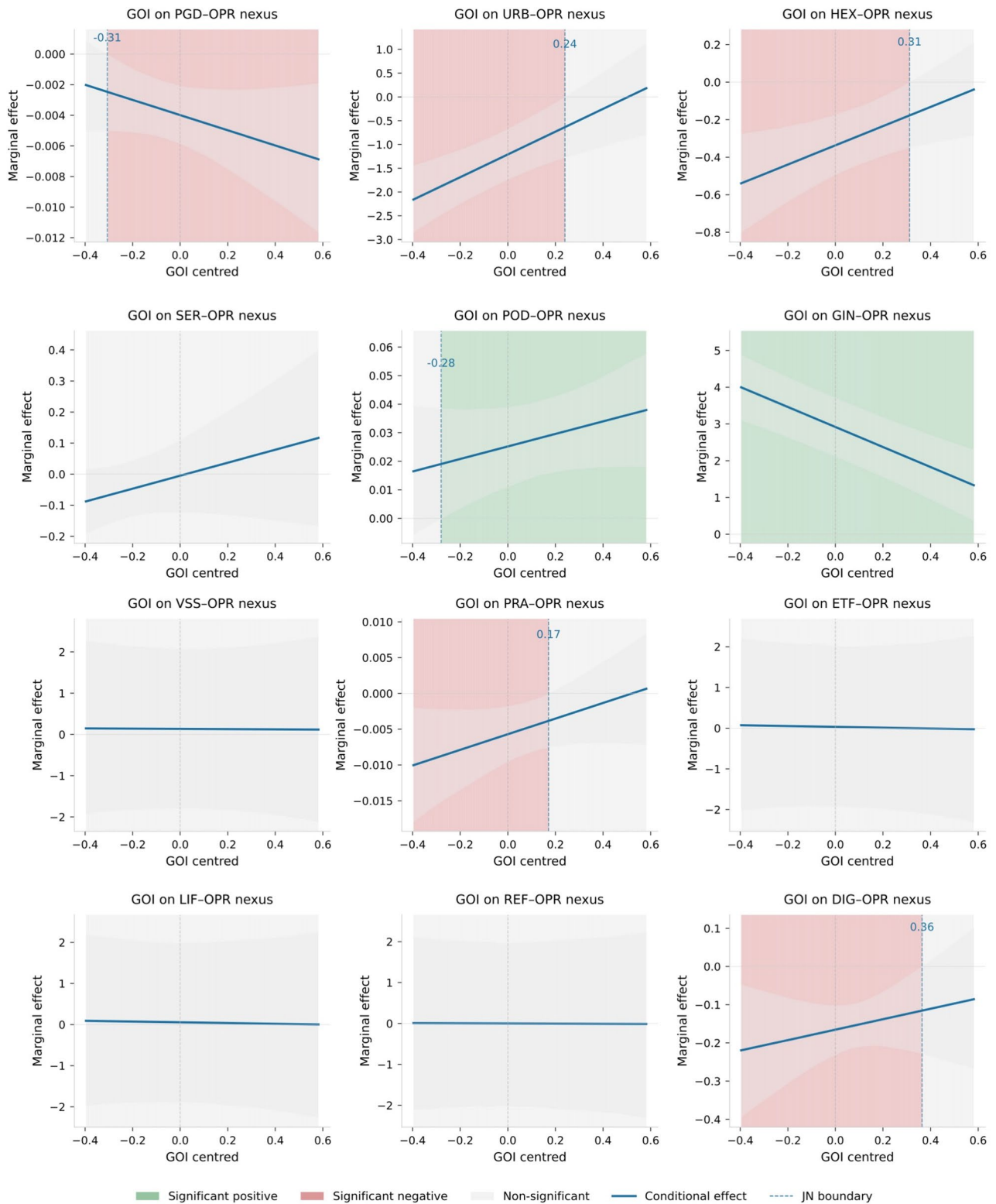


Fig. 4 Marginal effect plots for Model-III: OPR

Table 5 Robustness test results for model-I: SAT

Variable	HLM Wide N(0, 5)			Bayesian GLM			Bayesian structural IV		
	Coeff	Crl Lower	Crl Upper	Coeff	Crl Lower	Crl Upper	Coeff	Crl Lower	Crl Upper
PGD	0.001	0.00	0.00	0.002	0.00	0.00	0.029	−0.03	0.09
URB	1.337	1.06	1.64	−1.801	−2.39	−1.23	1.036	0.10	1.86
HEX	0.158	0.05	0.26	0.363	0.19	0.55	−0.562	−2.14	0.99
SER	0.027	−0.03	0.08	−0.099	−0.37	0.20	−2.791	−8.41	2.85
POD	0.009	0.00	0.02	0.016	0.01	0.02	0.009	0.00	0.02
GIN	0.930	0.31	1.49	2.182	1.42	2.92	2.010	0.33	3.74
VSS	−2.583	−11.98	7.17	−0.350	−5.24	4.56	−2.169	−12.05	7.72
PRA	0.002	0.00	0.00	0.012	0.00	0.03	−0.035	−0.11	0.04
GOI	−2.619	−8.24	3.30	−1.521	−6.18	3.50	0.276	−5.89	6.56
ETF	−0.225	−9.93	9.63	1.755	−3.04	6.38	−0.229	−9.88	9.60
LIF	−0.669	−10.27	9.15	−6.125	−11.26	−1.39	−1.113	−11.01	8.86
REF	−0.129	−9.87	9.49	0.878	−4.33	5.61	−0.304	−10.10	9.42
DIG	0.093	0.06	0.13	0.272	0.11	0.45	0.268	−0.09	0.71
Endogeneity (λ)							−0.029	−0.09	0.03
Model diagnostics									
SD (intercept)	41.231	23.85	75.39				32.500	22.10	48.49
σ (residual)	3.216	2.87	3.57	21.383	19.11	23.94	2.958	2.63	3.33
R-hat	1.000			1.000			1.020		
Bulk ESS	7026			9443			1393		
Chains	4			4			4		
Iterations	4000			4000			8000		

Note: Entries are posterior means with 95% credible intervals (Crl). s is the residual standard deviation; SD (intercept) reflects cross-country variation in intercepts. R-hat indicates satisfactory MCMC convergence; Bulk ESS reports effective sample size

Table 6 Robustness test results for Model-II: WAT

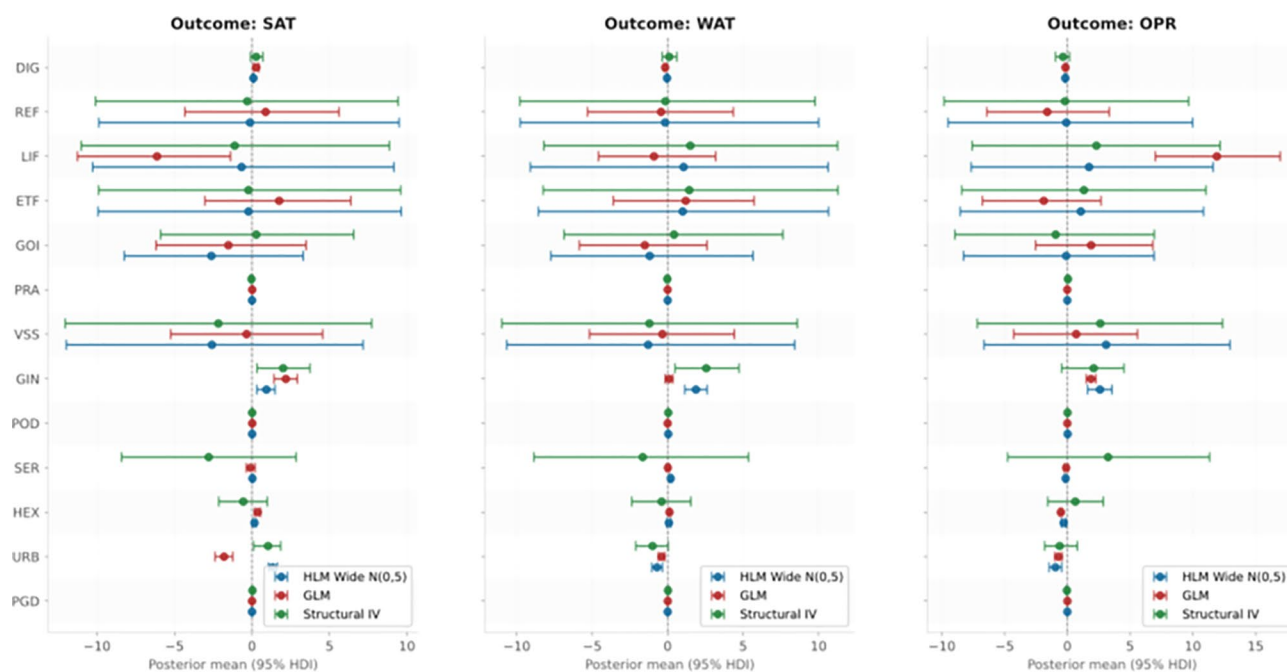
Variable	HLM Wide N(0, 5)			Bayesian GLM			Bayesian structural IV		
	Coeff	Crl Lower	Crl Upper	Coeff	Crl Lower	Crl Upper	Coeff	Crl Lower	Crl Upper
PGD	0.001	0.00	0.00	0.007	0.01	0.01	0.019	−0.05	0.09
URB	−0.705	−1.06	−0.34	−0.405	−0.61	−0.20	−1.003	−2.12	0.04
HEX	0.080	−0.06	0.21	0.117	0.05	0.19	−0.399	−2.38	1.53
SER	0.201	0.13	0.27	0.011	−0.09	0.11	−1.652	−8.86	5.35
POD	0.036	0.03	0.05	0.000	0.00	0.01	0.037	0.02	0.05
GIN	1.872	1.13	2.62	0.093	−0.18	0.35	2.555	0.48	4.74
VSS	−1.293	−10.66	8.41	−0.347	−5.18	4.42	−1.210	−11.00	8.59
PRA	0.001	0.00	0.00	−0.002	−0.01	0.00	−0.023	−0.12	0.07
GOI	−1.191	−7.74	5.66	−1.514	−5.86	2.61	0.422	−6.86	7.63
ETF	0.997	−8.56	10.66	1.195	−3.60	5.73	1.429	−8.25	11.28
LIF	1.053	−9.10	10.64	−0.906	−4.59	3.19	1.499	−8.20	11.26
REF	−0.170	−9.77	10.01	−0.435	−5.31	4.36	−0.152	−9.79	9.77
DIG	−0.046	−0.10	0.00	−0.168	−0.23	−0.11	0.098	−0.36	0.61
Endogeneity (λ)							−0.019	−0.09	0.05
Model diagnostics									
SD (intercept)	33.542	19.00	59.71				28.059	17.41	44.05
σ (residual)	4.254	3.79	4.72	7.010	6.27	7.78	4.169	3.72	4.69
R-hat	1.000			1.000			1.020		
Bulk ESS	11128			7722			1489		
Chains	4			4			4		
Iterations	4000			4000			8000		

Note: Entries are posterior means with 95% credible intervals (Crl). s is the residual standard deviation; SD (intercept) reflects cross-country variation in intercepts. R-hat indicates satisfactory MCMC convergence; Bulk ESS reports effective sample size

Table 7 Robustness test results for Model-III: OPR

Variable	HLM Wide N(0, 5)			Bayesian GLM			Bayesian structural IV		
	Coeff	Cri Lower	Cri Upper	Coeff	Cri Lower	Cri Upper	Coeff	Cri Lower	Cri Upper
PGD	-0.002	0.00	0.00	0.003	0.00	0.00	-0.035	-0.12	0.05
URB	-0.943	-1.45	-0.44	-0.703	-1.00	-0.41	-0.614	-1.81	0.80
HEX	-0.292	-0.47	-0.12	-0.517	-0.62	-0.42	0.634	-1.56	2.86
SER	-0.140	-0.24	-0.04	-0.100	-0.24	0.04	3.244	-4.76	11.33
POD	0.018	0.00	0.03	-0.003	-0.01	0.00	0.011	-0.01	0.03
GIN	2.599	1.62	3.55	1.888	1.51	2.27	2.098	-0.45	4.52
VSS	3.087	-6.64	12.96	0.707	-4.28	5.59	2.624	-7.18	12.37
PRA	-0.005	-0.01	0.00	-0.009	-0.02	0.00	0.040	-0.06	0.15
GOI	-0.096	-8.27	6.92	1.891	-2.53	6.82	-0.932	-8.96	6.93
ETF	1.073	-8.55	10.86	-1.879	-6.77	2.69	1.322	-8.42	11.05
LIF	1.730	-7.66	11.62	11.912	7.02	16.96	2.315	-7.58	12.17
REF	-0.090	-9.50	9.98	-1.602	-6.41	3.35	-0.187	-9.82	9.68
DIG	-0.167	-0.24	-0.10	-0.152	-0.24	-0.06	-0.340	-0.95	0.18
Endogeneity (λ)							0.034	-0.05	0.12
Model diagnostics									
SD (intercept)	28.740	15.37	51.82				23.092	13.43	38.01
σ (residual)	5.848	5.23	6.55	10.348	9.11	11.64	5.309	4.73	5.98
R-hat	1.000			1.000			1.010		
Bulk ESS	7634			8559			1984		
Chains	4			4			4		
Iterations	4000			4000			8000		

Note: Entries are posterior means with 95% credible intervals (Cri). σ is the residual standard deviation; SD (intercept) reflects cross-country variation in intercepts. R-hat indicates satisfactory MCMC convergence; Bulk ESS reports effective sample size

**Fig. 5** Forest plot of comparative analysis

governance, socioeconomic drivers, and WASH outcomes. The convergence of findings across prior distributions and model specifications strengthens confidence in the validity and stability of the reported conclusions.

Discussion

This study reveals that the drivers of WASH outcomes in South Asia are not uniformly direct but are conditional and outcome-specific, with governance quality acting as a selective moderator rather than a universal driver. The

finding that per capita GDP lacks a robust direct effect on WASH outcomes challenges purely economic determinism and aligns with institutional economics theory [38, 104]. These results also align with empirical studies by Coffey et al. [51] and Hathi et al. [75], who demonstrated that economic gains and infrastructure expansion alone fail to eliminate open defecation due to behavioural and governance constraints. However, our interaction results refine this interpretation. Rather than universally amplifying all drivers, governance selectively enhances the translation of economic capacity into improved sanitation and water access, while exhibiting no credible moderating role for open defecation. This explains the coexistence of high-growth yet low-WASH-access regions and reinforces a critical policy implication: economic development alone is insufficient without institutional mechanisms that enable effective targeting, coordination, and service delivery. These findings are in line with studies by Asongu and Odhiambo [105], Abdou et al. [49], Azimi et al. [106], Acheampong et al. [42], de Almeida et al. [107], and Balach and Law [108], who highlight that governance effectiveness amplifies the developmental dividends of economic expansion.

The divergent effects of urbanisation further illuminate this complex interplay. The association of higher urbanisation with improved sanitation and reduced open defecation, but lower water access, encapsulates the dual nature of urban transition: economies of scale in sanitation infrastructure versus the overwhelming strain on water resources and the proliferation of informal settlements. Crucially, our results show that governance is a key factor that tips this balance. Under strong governance, the benefits of density are harnessed while its liabilities are managed, transforming urbanisation from a liability into an opportunity for service delivery. This offers a powerful rebuttal to deterministic views of urban crisis and highlights the need for integrated urban governance strategies. These results align with prior research by McGranahan [40], McDonald et al. [53], Satterthwaite [109], Azeez et al. [55], Danelon et al. [57], and Azimi et al. [110], who show that the environmental outcomes of urbanisation depend heavily on institutional capacity and planning systems.

The consistent positive effect of public health expenditures confirms the vital role of fiscal commitment, particularly for sanitation and reductions in open defecation. However, the interaction results reveal a more complex dynamic than initially assumed. Governance appears to weaken the effectiveness of health expenditures for sanitation and water access, while strengthening its effect on open defecation. While this partially validates the concept of absorptive capacity [111], where the returns on public spending are contingent on the institutional environment's ability to implement spending efficiently and

target resources effectively, it suggests that institutional environments influence how resources are allocated and utilised across different WASH outcomes in the region. These findings are consistent with existing literature such as that by Booth [112], Pritchett et al. [113], Baldacci et al. [114], Hutton and Varughese [115], Hunter et al. [116], and Chaitkin et al. [117].

The results for the school enrolment ratio present a similarly complex picture. While its direct effects—improving water access and reducing open defecation—are consistent with established evidence linking education to hygiene behaviour and public health awareness [10, 58–60, 118, 119], governance does not uniformly enhance these effects. Instead, stronger governance appears to weaken the influence of education on sanitation and water access, with no credible moderating role on the SER-OPR nexus. This suggests that education operates through behavioural and community-level pathways that may not be fully captured or amplified by macro-level institutional quality, highlighting the need for integrated approaches that combine education with targeted behavioural interventions.

A particularly striking result is the positive association between income inequality and WASH outcomes. This pattern, observed across several empirical contexts, reflects a transitional dynamic in which early infrastructure expansion disproportionately benefits higher-income or urban groups, inflating aggregate coverage while the persistent exclusion of the poor remains hidden beneath the averages [61–63]. Consistent with Kuznets-style development sequences [120], this dynamic points to deeper structural and institutional mechanisms. Inequality signals the spatial and sectoral concentration of development, where WASH improvements tend to follow investments in economically dynamic regions, urban centres, or prioritised areas. In such contexts, aggregate coverage rises alongside persistent deprivation—a pattern echoed in recent works by Acheampong et al. [42] and Asongu and Odhiambo [105], who show that growth-linked improvements in WASH access may mask deep intra-national disparities. Crucially, governance quality moderates this relationship, strengthening improvements in sanitation and water access while reducing open defecation. Yet stronger governance does not necessarily imply more equitable outcomes; institutional capacity may facilitate more efficient service expansion without guaranteeing inclusive distribution. This aligns with evidence from Sahn and Younger [121] and Booth [112] indicating that, absent pro-poor targeting and redistributive policy frameworks—measures that extend beyond broad governance indicators—the benefits of public investment fail to trickle down equitably. Headline progress, in such cases, may entrench rather than reduce inequality.

The null results for supply pressure and shocks—operationalised through the vulnerability sector score and precipitation anomalies—are themselves revealing. They suggest that medium-term WASH outcomes in South Asia are shaped less by short-term climatic vulnerability than by structural and institutional conditions. This finding aligns with prior works demonstrating that governance capacity, infrastructure robustness, and socio-economic resilience mediate how environmental stress translates into service outcomes [122–124]. Climate-related factors, in this view, operate indirectly through economic and social channels rather than exerting immediate direct effects. Therefore, institutional capacity—not merely climate exposure—emerges as a key determinant of cross-regional variation in WASH vulnerability, a pattern echoed in studies by Jepson et al. [125] and Cadman and Maraseni [126]. Importantly, these null findings do not diminish the threat of climate change. Instead, they clarify a critical policy hierarchy: building resilient WASH systems requires first strengthening the governance and economic structures through which risks are mediated.

The cultural variables (ETF, LIF, REF) do not exhibit credible effects in either direct or moderating specifications. This suggests that, at the macro level, these dimensions may not adequately drive WASH outcomes once structural and institutional factors are accounted for. However, this does not imply that cultural factors are irrelevant; rather, their influence may operate through micro-level behavioural mechanisms not captured in cross-country aggregate data. This interpretation aligns with the Integrated Behavioural Model (IBM) of Dreibeilbis et al. [127] for WASH, which conceptualises cultural values and norms as operating across multiple levels—from structural and community contexts down to habitual and individual factors—all of which shape sustained WASH practices in infrastructure-constrained settings [128]. In contrast, digitalisation emerges as a robust determinant. Its positive association with sanitation and negative association with open defecation indicates that digital technologies enhance coordination, monitoring, and behavioural change mechanisms. However, governance negatively moderates this relationship, suggesting that digitalisation may function as a substitute for weaker institutional environments rather than a complement to strong governance.

Finally, the most profound finding is the consistent role of governance quality as a key moderator. The absence of a significant unconditional direct effect does not diminish its importance; its power lies not in direct provision but in its ability to enable, amplify, and redirect the impact of other sources. This provides strong, empirical confirmation of the institutionalist paradigm within development studies, positioning effective governance as

a critical mechanism without which economic growth, urbanisation, and fiscal capacity fail to achieve their full potential in improving human well-being. These findings align with Andrews [129], Joshi et al. [130], Halleröd et al. [131], Qian [132], Sachs et al. [133], and Azimi et al. [106], who collectively demonstrate that governance effectiveness and institutional capacity determine the translation of economic and fiscal inputs into sustainable development practices.

More broadly, our findings carry important implications for global health systems and WASH programme design. They suggest that interventions should move beyond infrastructure provision toward integrated approaches that combine governance reform, targeted investments, and technological innovation. The absence of a direct governance effect indicates that improving institutional quality in aggregate is insufficient; rather governance reforms must be sector-specific and tightly aligned with service delivery mechanisms. This nuanced view of governance reflects the broader pattern of hypothesis testing. While health expenditures and education emerge as strongly supported determinants—reinforcing the foundational importance of human capital and fiscal commitment—direct effects for economic capacity, environmental shocks, and governance are not corroborated. Yet governance functions as a significant moderating factor, confirming that institutional quality conditions the effectiveness of key structural drivers. Together, these results suggest that WASH programming requires not only investing in the right sectors but also ensuring that institutional arrangements can translate those investments into sustainable outcomes.

Conclusions

This study investigated the determinants of WASH outcomes across eight South Asian countries over the period 2000–2024, a region characterised by persistent service deficits and profound structural heterogeneity. To address the limitations of classical panel and frequentist estimators, the study employs a Bayesian Hierarchical Linear Model (HLM), validated through alternative prior distributions, pooled Bayesian generalised linear models (GLM), and a Bayesian structural instrumental variable method. A key methodological contribution is the construction of a comprehensive governance quality index from multiple administrative and political indicators via an autoencoder neural network tailored to WASH dynamics in South Asia. The results demonstrate that structural and demographic variables exert heterogeneous and outcome-specific effects on WASH indicators, while governance quality primarily operates as a moderating force. The direct effects reveal a complex landscape; urbanisation has a dual influence, improving sanitation access while straining water availability and

reducing open defecation. Health expenditures contribute positively to sanitation and the reduction in open defecation but show no credible effect on water access, highlighting uneven sectoral impacts of public spending. School enrolment improves water access and reduces open defecation, reinforcing the importance of education-driven behavioural channels. Population density shows a small but positive effect. In contrast, economic capacity, sectoral vulnerability, precipitation shocks, cultural affiliations, and governance quality do not exhibit direct effects, challenging simplistic development narratives. The moderated models help to resolve most of this puzzle: when modelled interactively, governance quality significantly amplifies the positive effects of economic growth, urbanisation, digitalisation, and inequality. It weakens the influence of health expenditures and education on certain outcomes, indicating that institutional quality does not always enhance the effectiveness of structural inputs. These findings suggest that governance operates through complex and context-dependent channels rather than as a universally beneficial force.

Policy implications

Our findings highlight several important policy implications. They underscore the need to channel resources through capable institutions, as growth and sectoral spending translate into WASH improvements only when allocation, oversight, and execution are credible. In the context of South Asia, this requires strengthening core state capacity through digital public financial management systems, transparent procurement platforms, and decentralised monitoring frameworks that enable real-time tracking of WASH investments. Strengthening procurement integrity, budget transparency, and delivery-chain accountability can substantially magnify the returns to existing fiscal envelopes. These reforms are directly aligned with SDG 16, which underpins the effectiveness of sectoral investments in SDG 6. At the same time, managing the urbanisation trade-off remains crucial: while urban growth can accelerate sanitation coverage, it also strains water systems and affects progress in reducing open defecation. Given the rapid expansion of informal settlements across South Asian cities, WASH strategies must prioritise inclusive urban planning that integrates slum upgrading, decentralised sanitation systems, and equitable service access. Urban WASH strategies must therefore combine infrastructure expansion with demand-side management, climate-resilient water systems, and pro-poor connection policies, implemented under clear accountability regimes. These priorities are closely linked to SDG 11, particularly in ensuring safe, inclusive, and resilient urban service delivery.

Furthermore, education investments generate important social co-benefits by enhancing uptake and sustained

use of WASH services. However, the results indicate that education alone is insufficient without effective institutional delivery; therefore, policies must integrate school-based WASH programmes, community-level behavioural interventions, and local governance mechanisms to ensure knowledge translates into sustained behavioural change. The findings also highlight the emerging role of digitalisation as a direct driver of improved WASH outcomes. This suggests that governments in South Asia should invest in digital infrastructure for service delivery, including smart metering, geospatial monitoring of water access, and data-driven targeting of underserved populations. Finally, because governance primarily modulates the effects of structural drivers, institutionalising adaptive governance is essential for converting structural change into durable improvements in access. This can be operationalised through three core channels: (a) improving allocative efficiency via transparent and targeted budgeting, (b) strengthening implementation integrity through accountable procurement and digital monitoring systems, and (c) ensuring sustainability through continuous monitoring, feedback loops between communities and service providers, strengthening local government capacity, and embedding performance-based accountability systems within WASH programmes. In the South Asian context, where administrative capacity varies significantly across regions, such adaptive governance frameworks are critical for ensuring that policies remain responsive to local conditions.

Limitations

This study has several limitations. First, reliance on secondary data may introduce comparability issues, as coverage and measurement precision vary over the 2000–2024 period. Second, the country-level analysis, while suitable for cross-national inference, abstracts from sub-national inequalities and service-network heterogeneity that are critical for understanding last-mile access. Third, although the Bayesian HLM framework effectively captures cross-country heterogeneity through random intercepts, it does not establish causality and may absorb unobserved structural factors—such as differences in natural resource endowments—without explicitly identifying their effects. Future studies should address these limitations by integrating more granular sub-national data to capture within-country disparities in WASH outcomes and institutional performance. Disaggregated analysis would allow for the examination of rural-urban divides, gendered access gaps, and local governance dynamics. In addition, methodological innovation is needed: causal inference techniques such as natural experiments, instrumental variable designs, and staggered policy adoption models could provide stronger identification of mechanisms.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12992-026-01219-7>.

Supplementary Material 1

Author contributions

Mohammad Naim Azimi: Original draft, data collection, computation, and analysis. Mohammad Mafizur Rahman: Editing, validation of analysis, and supervision.

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Data availability

All datasets used will be available upon request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interest

The authors declare no competing interests.

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